KKT KOLBE HERMES RGBW Hood

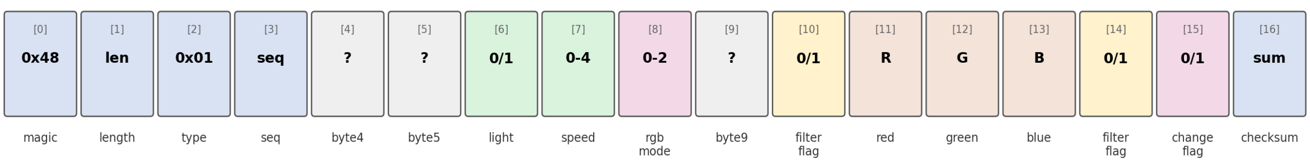
Hekr Protocol Reference — Address Space Map

This document maps every address space in the raw-hex protocol used to control the cooker hood over its Hekr/Wisen cloud connection (hub.hekreu.me:83), reverse-engineered through packet capture, systematic command sweeping, official firmware SDK cross-reference, and community protocol documentation. It covers two distinct address spaces: the **byte layout** within each frame, and the **command ID space** that selects what a frame does.

1. Status Frame Byte Layout

Sent by the device to the cloud (and relayed to Home Assistant) roughly every second, and immediately after any state change. Frame type byte is always 0x01 for this direction.

Status Frame Byte Layout (device -> cloud, e.g. 481101E501000000020000398C69000171)



Offset	Field	Description
0	**Magic byte**	Constant 0x48. Marks the start of every frame, both directions.
1	**Length**	Total frame length in bytes, including this byte and the final checksum.
2	**Frame type**	0x01 for a status report. (0x02 is used for the command direction — see Section 2.)
3	**Sequence**	Rolling counter incremented by the device on every frame it sends. Not used for authentication.
4	**byte4**	Unidentified. Observed as 0x01 in nearly every capture; role unknown.
5	**byte5**	Unidentified. Occasionally changes alongside light/speed during panel use; exact role unclear.
6	**Light (white)**	0 = off, 1 = on. The white light channel, independent of the RGB colour. Settable via Command ID 0x03.
7	**Fan speed**	0 = off, 1–4 = speed levels. Settable via Command ID 0x04.
8	**RGB mode flag**	0 = never engaged (seen only on first-ever connection). 1 = off, but colour value is remembered. 2 = on. Fully solved 2026-08-13: cleanly and reliably togglable via cmdId 0x03 (the same “Light” command used for the white channel), no fan or power-cycle required — provided the current colour was stored via mode=0x00 rather than mode=0x02 (see Section 3, Command ID 0x07).
9	**byte9**	Unidentified. Always 0x00 in every capture so far.
10	**Filter reminder flag**	0 = no warning, 1 = filter needs cleaning. Confirmed 2026-08-13 against a real, organically-triggered filter warning. Mirrors byte 14 exactly — both flip together. Previously listed as unidentified/always-0x00; only ever differs from 0 once the panel's actual filter indicator is lit.
11	**Red**	0–255. Confirmed by cycling all 9 physical panel colour presets and cross-referencing each against this byte.
12	**Green**	0–255. Confirmed as above.
13	**Blue**	0–255. Confirmed as above.

Offset	Field	Description
14	**Filter reminder flag**	Mirrors byte 10 (see above). Reason for the duplication unknown — possibly a redundancy/checksum-adjacent design choice, possibly two related-but-distinct conditions that happen to always coincide in testing so far.
15	**Change flag**	Transient “just changed” pulse: briefly 1 right after any state change, then reverts to 0 on the next heartbeat. Ignore when reading steady-state values.
16	**Checksum**	Sum of every preceding byte in the frame, modulo 256 (simple byte-sum, no CRC polynomial). Confirmed identical across three independent open-source implementations of this protocol.

2. Command Frame Shapes (app/bridge → device)

Commands sent to the device follow the same magic-byte/length/type/sequence header, but with frame type 0x02, and a payload shape that depends on the Command ID. Four shapes are confirmed:

Simple command (Power 0x02, Light 0x03, Speed 0x04, Timer 0x05)

`example: 480702D70301 2C   ->   Light ON`

0x48	**07**	**02**	**seq**	**cmdId**	**value**	**sum**
magic	length	type	seq	cmd ID	value	checksum

Colour command (cmdId 0x07 only)

`example: 480A02E80702398C6973   ->   set colour to #398C69, RGB on`

0x48	**0A**	**02**	**seq**	**07**	**mode**	**R**	**G**	**B**	**sum**
magic	length	type	seq	cmd ID	mode	red	green	blue	checksum

Timer command (cmdId 0x05)

`example: 480702D80501 2F   ->   auto-off timer, duration unit 1`

0x48	**07**	**02**	**seq**	**05**	**units**	**sum**
magic	length	type	seq	cmd ID	duration	checksum

Clock-set command (cmdId 0x08) — confirmed 2026-08-11

`example: 48090204080F270095   ->   set clock to 15:39:00`

0x48	**09**	**02**	**seq**	**08**	**hour**	**min**	**sec**	**sum**
magic	length	type	seq	cmd ID	hour	minute	second	checksum

Note the colour command's extra **mode** byte before R/G/B — without it, the device silently misinterprets the frame; this was the key discovery that unlocked reliable colour control in the first place. The mode byte's own value matters just as much: **mode=0x00** stores a colour while leaving the device genuinely off, while **mode=0x02** stores and turns it on but permanently “stickies” the RGB status flag — the second key discovery, which finally unlocked genuine fan-free on/off (see Section 4).

3. Command ID Address Space

Command ID occupies byte offset 4 of a command frame and selects which function the payload targets. Every value from 0x02 through 0x14 has been tested; results below.

ID	Function	Status	Notes
0x02	Power	Working	Master fan+light power. Value byte: 0/1.
0x03	Light (white)	Working	White light only, independent of RGB. Value byte: 0/1.
0x04	Fan speed	Working	Value byte: 0–4.

ID	Function	Status	Notes
0x05	Auto-off timer	Working	Value byte sets countdown duration (units unconfirmed, roughly seconds-to-minutes). Panel-cancellable. On completion, only resets its own duration byte (offset 9) — confirmed to have zero effect on the RGB mode flag (offset 8) or any other field. A value of 0x02 produces a ~90-second countdown; retested cleanly on 2026-08-11 after an earlier ambiguous log (from 2026-07-28) raised a false alarm suggesting it might toggle the RGB flag directly — that turned out to be coincidental timing with a manual panel press, not a real effect.
0x06	Filter/clean reminder reset	Working, confirmed	Value byte 0x00. Solved 2026-08-13: earlier extensive testing (bare, 1-byte, extended 2-byte payloads) consistently found no response, which turned out to be because a filter-reset command has no observable effect while no filter warning is active. Retested against a real, organically-triggered warning and confirmed immediately — response arrived in under one second and cleared both status bytes 10 and 14, plus the physical panel icon.
0x07	RGB colour	Fully solved, 2026-08-13	Requires 4-byte payload: mode + R + G + B. Full mode semantics finally confirmed: **mode=0x00** stores the given colour and leaves the device genuinely off (status byte 8 -> 1, matching the physical panel indicator). **mode=0x02** stores the colour and turns it on (status byte 8 -> 2) — but critically, using mode=0x02 to change colour permanently “stickies” the RGB flag, breaking plain Light on/off (cmdId 0x03) until a full power-cycle reset. mode=0x03 is accepted and behaves identically to 0x02. mode=0x01 (originally assumed “off” by symmetry) is never acknowledged by the device, in any context, across dozens of tests — confirmed non-functional. The correct colour-change sequence is mode=0x00 (store) followed immediately by cmdId 0x03 value 1 (display it) — see Section 4.
0x08	Clock (time-of-day)	Working, confirmed	Two distinct uses confirmed. (1) Simple 1-byte values accidentally adjust the wall-clock display (early discovery, required a manual reset). (2) Proper 3-byte payload — [hour, minute, second], with a genuinely computed checksum, not the hardcoded 0x00 a reference implementation used — reliably sets the clock to an exact time. Confirmed working 2026-08-11 and built into the add-on as a “Sync Clock” button. The clock is never reported back in the status frame — this command is fire-and-forget, verified visually only.
0x09	RGB on/off (per community docs)	Documented, not functional	Identified via an independent open-source adapter's protocol table (cmdId 9 = “RGB_Sw”). That project's own author marked it “not implemented” after being unable to get a response. Exhaustively retested on this hood: bare, 1-byte, extended 2-byte, the reference adapter's exact literal format, a 3-byte colour-shaped payload, and the full 4-byte [mode,R,G,B] shape that unlocked cmdId 0x07 — 15 distinct variants total, tested with the RGB light confirmed genuinely on beforehand (to rule out silence-because-nothing-to-turn-off). No response in any case.
0x0A–0x0D	—	No response	Tested bare and with 1-byte values (0/1). Silent every time.
0x0E–0x14	—	No response	Tested bare, with 1-byte values, and (0x0E–0x10 only) with a 3-byte RGB-shaped payload. Silent every time.
various	Frame-type byte variants	No effect	Tested 0x00, 0x01, 0x03, 0x04 in place of the usual 0x02 frame-type byte, paired with cmdId 0x07. No response from any.

4. How This Maps to the Home Assistant Add-on

The `hekr_bridge` add-on builds and parses these exact frames. Practical behaviour, given everything above:

- **Power / Light / Fan Speed** entities map directly to Command IDs 0x02/0x03/0x04 — fully reliable.
- **RGB Light** entity: genuine, fan-free on/off fully solved 2026-08-13. Setting a new colour sends **mode=0x00** (store, stay genuinely off) immediately followed by **cmdId 0x03 value 1** (display it) — this keeps the RGB flag out of the "sticky" state that mode=0x02 causes. From then on, plain on/off both route through cmdId 0x03 alone; the device remembers its own last colour internally, so no colour needs to be resent on a plain "turn on". The black-colour workaround and the fan-blip reset trick from earlier in the investigation are no longer used anywhere in the add-on.
- **On/off status** shown in Home Assistant now reads directly from status byte 8 (the RGB flag) rather than inferring it from the colour value — safe to do now that byte 8 is reliably and predictably toggled by cmdId 0x03, including correctly reflecting changes made from the physical panel too.
- **Sync Clock** button (added 2026-08-11) uses Command ID 0x08 to set the hood's clock to the add-on's current system time, or an explicit override time if published to the same topic — useful if the add-on container's timezone doesn't match the hood's physical location.
- **Filter Needs Cleaning** binary sensor and **Reset Filter Reminder** button (added 2026-08-13) read and clear status bytes 10/14 via Command ID 0x06 — confirmed working against a real filter warning, closing out what had been an open investigation since the project began.

Sources cross-referenced during this investigation: direct packet capture and command sweeping against this device; Hekr's official HEKR-MCU-SDK reference firmware; the ioBroker.hekr community adapter; the hekrapi-python protocol library; and the decompiled Wisen Android application.

Revised 2026-08-13 (later): genuine RGB off fully solved. The missing piece was the colour-set command's mode byte itself — mode=0x00 stores a colour without engaging the RGB flag, unlike mode=0x02 which was being used throughout the project up to this point. No project limitations remain outstanding. Revised 2026-08-13 (earlier): filter/clean reminder fully solved (indicator bytes and reset command both confirmed against a real warning). Revised 2026-08-11: added confirmed clock-set command, retested cmdId 0x05/0x06 findings.